

Firestore with Dart

Cloud NoSQL Database & Backend-as-a-Service

What is Firebase?

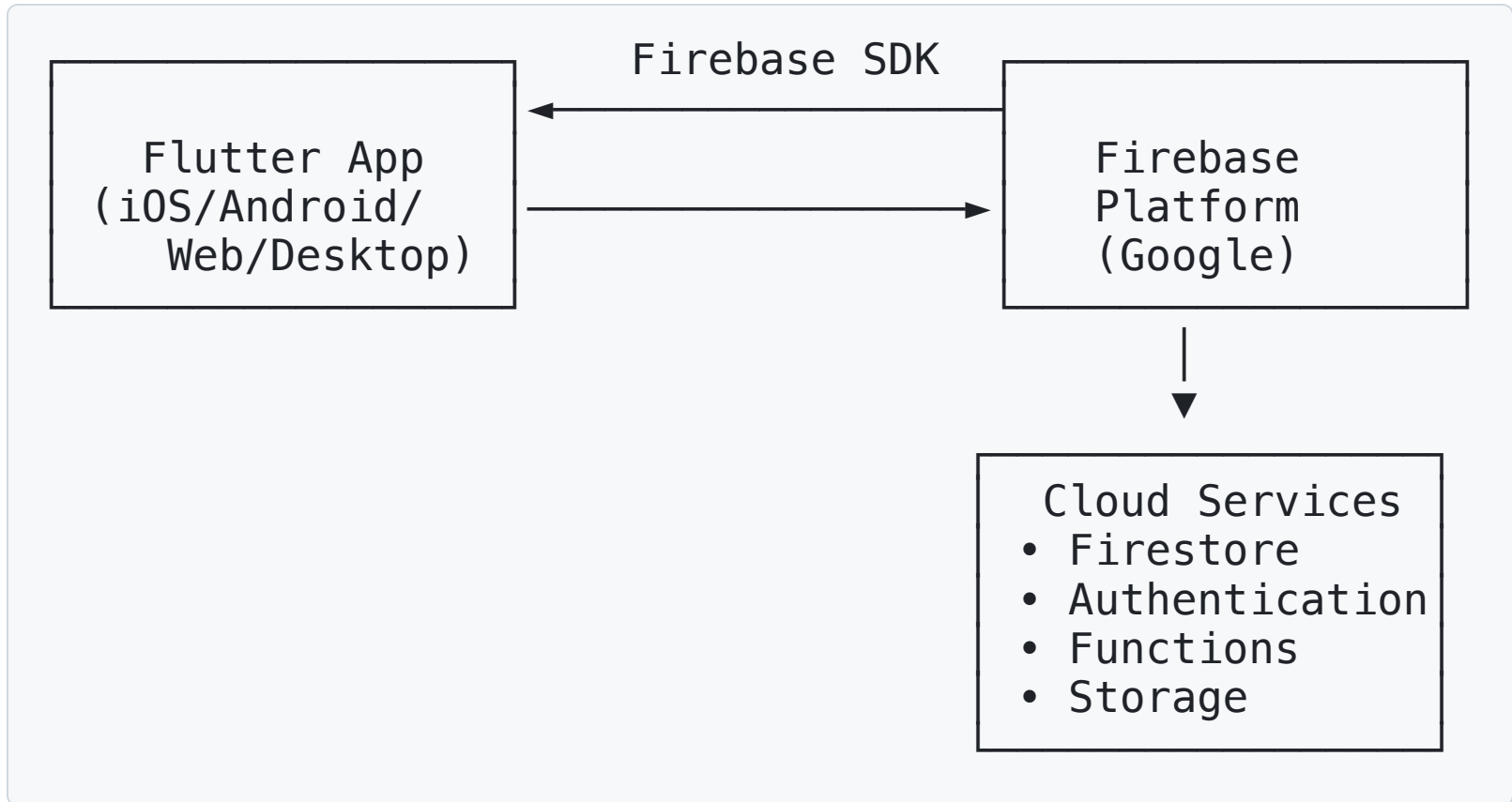
- **Google's Backend-as-a-Service** platform
- **Cloud Firestore** - NoSQL document database
- **Real-time synchronization** across devices
- **Built-in authentication** and security
- **Cross-platform** - iOS, Android, Web, Desktop

Firestore Services:

- Firestore (Database)
- Authentication
- Cloud Functions
- Cloud Storage
- Analytics
- Hosting
- Push Notifications

Used by: WhatsApp, Spotify, Airbnb, The New York Times

Architecture Overview



Benefits:

- No server management required
- Automatic scaling and backup
- Real-time data synchronization
- Global edge network

foo project

- In this project, we use Firebase.
- Use "Firebase_Quick_Start_Guide.md" for registering and making Firebase/firestore project.
 - We assume that users already made the foobar project.

Firebase Project

- Make sure you installed firebase CLI tool.
- Get your Firebase "Project ID" for your Dart project.

```
> firebase projects:list
```

Preparing the list of your Firebase projects

Project Display Name	Project ID	Project Number	Resource Location ID
foobar	foobar-YOUR_ID	827133271343	[Not specified]

Project Setup

```
# pubspec.yaml
dependencies:
  firedart: ^0.9.8
```

Then download the dependencies.

```
dart pub get
```

Import the package.

```
import 'package:firedart/firedart.dart';
```


Firestore

- Using Firebase in Dart is working with **Cloud Firestore** - NoSQL document database
- Firebase works with **collections** and **documents** (not tables and rows)
- Firebase is **cloud-based** - works on web, mobile, and desktop
- Firebase provides **real-time updates** and **offline support**

Initialization

```
import 'package:firedart/firedart.dart';

Future<void> main() async {
  // Initialize Firestore
  // with your project ID (not Firebase app)
  Firestore.initialize("foobar-YOUR_ID");

  // Get Firestore instance
  final firestore = Firestore.instance;
  ...
}
```

CRUD

- Use `add()` to create new documents with auto-generated IDs
- Use `set()` to create/update documents with specific IDs
- Use `get()` to retrieve documents once
- Use `snapshots()` to listen for real-time changes

Create: `collection().add()`

- Creates document with auto-generated ID
- Safe: No ID conflicts

```
Document addedDoc = await firestore
    .collection('foo')
    .add(generateRandomData());
```

Create/Update: `collection().document().set()`

- Uses specific document ID
- Creates new or overwrites existing
- Risk: Can overwrite existing data

```
await FirebaseFirestore.instance
  .collection('foo')
  .document(foo.id)
  .set(foo.toMap());
```

Read Once: `collection().document().get()`

- Fetches document data one time
- Good for displaying current state
- Efficient: Single network request

```
Document retrievedDoc = await firestore
    .collection('foo')
    .document(foo.id)
    .get();
```

Read Live: `collection().snapshots()`

- Creates real-time stream of changes
- Updates automatically when data changes
- Cost: Continuous connection and billing

```
Stream<QuerySnapshot> stream = firestore  
    .collection('foo')  
    .snapshots();
```

Update Fields: `doc().update()`

- Updates specific fields only
- Preserves other existing fields
- Efficient: Only changes specified data

```
await firestore
    .collection('foo')
    .document(foo.id)
    .update({'bar': 21, 'foo': 'Data Science'});
```


Delete Document: `doc().delete()`

- Completely removes document
- Cannot be undone
- Risk: Permanent data loss

```
await FirebaseFirestore.instance
  .collection('foo')
  .document(foo.id)
  .delete();
```

foobar project

- In this project, we use foobar data model to make Dart firebase application.
- Compared to the foo project that aims to understand firebase operation in Dart, foobar project is well designed with OOP.

- FooBar data model

```
class FooBar {  
    // Document ID from Firebase  
    // (nullable for new documents)  
    final String? id;  
    final String foo;           // String field  
    final int bar;             // Integer field  
  
    /// Constructor with required fields  
    FooBar({  
        this.id,  
        required this.foo,  
        required this.bar,  
    });  
}
```

```

Map<String, dynamic> toMap() {
    return {
        'foo': foo,
        'bar': bar,
    };
}

static FooBar fromMap(Map<String, dynamic> map,
[String? documentId]) {
    return FooBar(
        id: documentId,
        foo: map['foo'] ?? '',
        bar: map['bar'] ?? 0,
    );
}

```

- String? documentId is an optional positional parameter.

```
FooBar copyWith({  
    String? id, String? foo, int? bar,  
}) {  
    return FooBar(  
        id: id ?? this.id,  
        foo: foo ?? this.foo,  
        bar: bar ?? this.bar,  
    );  
}
```

- Create a copy of this FooBar with some fields updated.
- Useful for update operations

Processing ID

- In our data model, we have the `id` , but we don't set this value, but Firebase automatically assigns the value.
- The retrieved doc from Firebase has id and map components.

```
Document doc = await firestore
    .collection('foo')
    .document(foo.id)
    .get();
```

We create a new Dart object from `doc.map` and `doc.id`.

```
static FooBar fromMap(Map<String, dynamic> map, [String? documentId]) {
    return FooBar(
        id: documentId,
        foo: map['foo'] ?? '',
        bar: map['bar'] ?? 0,
    );
}

FooBar foobar = FooBar.fromMap(doc.map, doc.id);
```

CRUD

Service Class for Firestore Operations

```
class FooBarCrudFirebase {  
    late Firestore _firestore;  
    final String _collectionName = 'foobars';  
  
    Future<void> initialize({String projectId = 'foobar-PROJECT'})  
    async {  
        try {  
            Firestore.initialize(projectId);  
            _firestore = Firestore.instance;  
        } catch (e) {  
            rethrow;  
        }  
    }  
}
```


CREATE: Add new student to Firestore

```
Future<FooBar?> create(FooBar foobar) async {  
  try {  
    Document doc = await _firestore  
      .collection(_collectionName)  
      .add(foobar.toMap());  
    // Return the FooBar with the new ID  
    return foobar.copyWith(id: doc.id);  
  } catch (e) {  
    return null;  
  }  
}
```

- Returned FooBar object has auto-generated ID.

READ: Get a single FooBar by ID

```
Future<FooBar?> read(String id) async {  
  try {  
    print('📖 Reading FooBar with ID: $id');  
    // Get document by ID  
    Document doc = await _firestore  
      .collection(_collectionName)  
      .document(id).get();  
    // Convert to FooBar object  
    FooBar foobar = FooBar.fromMap(doc.map, doc.id);  
    print('✅ FooBar retrieved: $foobar');  
    return foobar;  
  } catch (e) {  
    print('❌ Error reading FooBar: $e');  
    return null;  
  }  
}
```

READ: Get all FooBar documents

```
Future<List<FooBar>> readAll() async {
  try {
    print('📖 Reading all FooBar documents...;');
    // Get all documents from collection
    List<Document> docs = await _firestore
      .collection(_collectionName)
      .get();
    // Convert to FooBar objects
    List<FooBar> foobars = docs
      .map((doc) => FooBar.fromMap(doc.map, c.id))
      .toList();
    print('✅ Retrieved ${foobars.length} oBar documents');
    return foobars;
  } catch (e) {
    print('❌ Error reading all FooBars: $e');
    return [];
  }
}
```

READ: Query FooBar documents where bar value equals the given number

```
Future<List<FooBar>> readByBar(int barValue) async {
  try {
    print('🔍 Querying FooBars where bar = arValue');
    // Query documents with filter
    List<Document> docs = await _firestore
      .collection(_collectionName)
      .where('bar', isEqualTo: barValue)
      .get();
    // Convert to FooBar objects
    List<FooBar> foobars = docs
      .map((doc) => FooBar.fromMap(doc.map, c.id))
      .toList();
    print('✅ Found ${foobars.length} FooBars th bar = $barValue');
    return foobars;
  } catch (e) {
    print('❌ Error querying FooBars: $e');
    return [];
  }
}
```

UPDATE: Modify an existing FooBar document

```
Future<bool> update(String id, FooBar datedFoobar) async {  
  try {  
    print('✎ Updating FooBar with ID: $id');  
    print('    New data: $updatedFoobar');  
    // Update document  
    await _firestore  
      .collection(_collectionName)  
      .document(id)  
      .update(updatedFoobar.toMap());  
    print('✅ FooBar updated successfully');  
    return true;  
  } catch (e) {  
    print('❌ Error updating FooBar: $e');  
    return false;  
  }  
}
```

UPDATE: Partially update specific fields

```
Future<bool> updateFields(String id, p<String, dynamic> updates) async {  
  try {  
    print('🔄 Updating FooBar fields for ID: d');  
    print('  Updates: $updates');  
    // Update specific fields  
    await _firestore  
      .collection(_collectionName)  
      .document(id)  
      .update(updates);  
    print('✅ FooBar fields updated ccessfully');  
    return true;  
  } catch (e) {  
    print('❌ Error updating FooBar fields: ');  
    return false;  
  }  
}
```

DELETE: Remove a FooBar document

```
Future<bool> delete(String id) async {  
  try {  
    print('🗑️ Deleting FooBar with ID: $id');  
    // Delete document  
    await _firestore  
      .collection(_collectionName)  
      .document(id)  
      .delete();  
    print('✅ FooBar deleted successfully');  
    return true;  
  } catch (e) {  
    print('❌ Error deleting FooBar: $e');  
    return false;  
  }  
}
```

Utility functions

```
Future<int> count() async {
```

Firestore and Flutter

- To use Firestore with Flutter, we need to add more configuration files.
- Use "Firestore_Quick_Start_Guide.md" for detailed the installation and configuration.

Firebase CLI tools

One time installation.

```
dart pub global activate flutterfire_cli
```

For each Flutter project that uses Firebase, we need to configure to use Firebase.

```
flutterfire configure
```

pubspec.yaml

For flutter applications:

```
dependencies:  
  firebase_core: ^2.24.2  
  cloud_firestore: ^4.13.6
```

For web applications, we add:

```
# Add this for web support  
firebase_core_web: ^2.10.0  
cloud_firestore_web: ^3.8.10
```

Developing Flutter Applications + Firebase

1. Make sure "lib/firebase_options.dart" file is generated from `flutterfire configure`.
2. Add dependencies and initialization code.

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'dart:math';
import 'firebase_options.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();

  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );
  runApp(MyApp());
}
```

Web Applications

- Use "database/firebase/foobar_flutter_webapp" as an example.
- Update "web/index" for accessing Firebase from JavaScript.

Developing other platforms + Firebase

- Use "database/firebase/foobar_flutter_app" as an example.
- Make sure you use the correct OS version to support (mac/ios).
- Make sure the app can use network (mac/ios).

Databases

Use Case	IndexedDB	Firebase	SQLite	PocketBase
Browser-only apps	✓ Perfect	⚠ Overkill	✗ Not available	✗ Not available
Offline-first web	✓ Excellent	✓ Smart sync	✗ Not available	✗ No offline
Large data storage	✓ Good (250MB+)	⚠ Expensive	✓ Unlimited	⚠ Server dependent
Complex queries	✗ Limited	✓ Rich NoSQL	✓ Full SQL	✓ REST API
Real-time sync	✗ Manual	✓ Automatic	✗ Manual	✓ Built-in
Multi-device sync	✗ No	✓ Automatic	✗ Manual	✓ Automatic
Learning curve	⚠ Moderate	✓ Easy	✓ Simple	✓ Easy

Decision Framework

- **Choose PocketBase** for: Self-hosted real-time apps, educational projects, MVPs, data control`
- **Choose IndexedDB** for: Browser-only applications, offline-first web apps, client-side caching

Choose SQLite for: Single-user apps, offline-first, embedded applications

Choose Firebase for: Global scale, automatic scaling, rapid development without hosting

Firestore Limitations

Database Structure Limitations

- No complex queries or JOINS across collections
- Maximum document size: 1 MB
- Limited filtering (max 30 composite indexes)
- Denormalization required → data duplication

Performance Limitations

- Maximum sustained writes: 10,000/second per database
- Single document: 1 write/second sustained
- No server-side aggregations
- Limited offline query capabilities

Cost Limitations

- **Reads:** \$0.36 per 100K documents
- **Writes:** \$1.08 per 100K documents
- **Storage:** \$0.18/GB/month
- Bandwidth charges for large documents

Feature Limitations

- No transactions across multiple collections
- No stored procedures or triggers
- Limited local development tools
- Vendor lock-in with Google ecosystem

Firestore Offline Support

- **Local caching:** Data is stored on device even when offline
- **Automatic sync:** Changes are synced when connection is restored
- **Read operations:** Continue to work using cached data
- **Write operations:** Queued locally and replayed once online
- **Cross-platform:** Works on iOS, Android, and Web